

IoT Device Security Scanner

Mini Project - Cyber Security in Mobile, Cloud and IoT

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Div:D

Introduction

- IoT devices like routers, IP cameras, smart TVs and smart plugs are now common in every home.
- Many of these devices run insecure services such as Telnet or MQTT without encryption.
- Attackers use these open services to take control of devices. Example: the Mirai botnet used Telnet and default passwords.
- This project scans a local network and shows which devices have risky ports open.

Common IoT devices

Wi-Fi Router

IP Camera

Smart TV

Printer

Problem Statement and Objectives

Problem

Users usually do not know which devices are connected to their network or which of them have insecure services running. There is no simple way to check this.

Objectives

- Scan a local network
- Find connected devices
- Check open ports
- Identify IoT services
- Show the risk level
- Give recommendations
- Show results on a web page

Technologies Used

Python 3

Main language used for scanning (socket, subprocess, ip address modules)

Flask

Small Python web framework used to run the dashboard on localhost

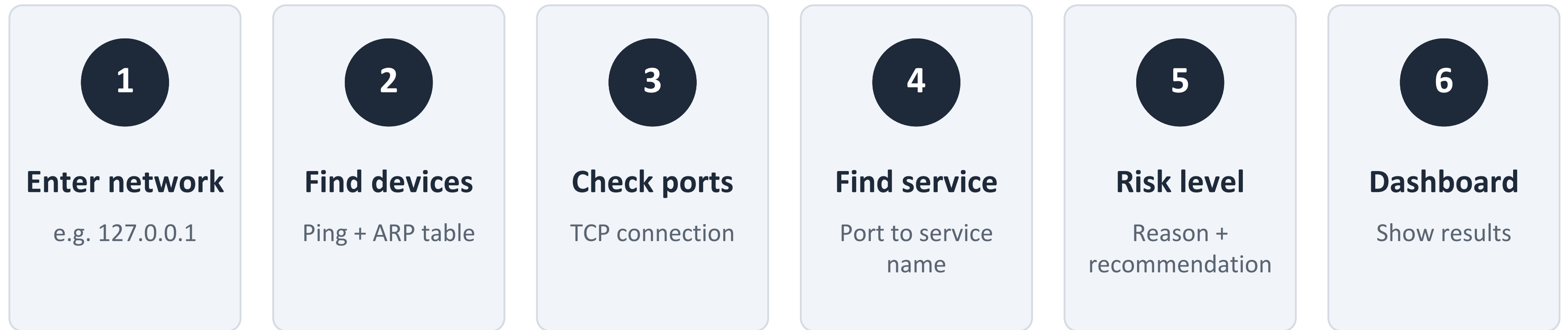
HTML and CSS

Structure and design of the dashboard page

JavaScript

Sends the scan request and shows the results in tables

Working of the Project



Only private networks are allowed. Public IP ranges are blocked by the program.

Ports Checked

Port	Service	Risk
21	FTP	High
22	SSH	Low
23	Telnet	High
80	HTTP	Medium
443	HTTPS	Low

Port	Service	Risk
554	RTSP	Medium
1883	MQTT	High
2323	Telnet (alternate)	High
8080	HTTP (alternate)	Medium
8883	MQTT over TLS	Low

Each open port is matched with its service, risk level, reason and recommendation.

Risk Levels

Critical

Two or more High risk services open on the same device

High

Unencrypted remote access or messaging - Telnet, FTP, MQTT

Medium

Unencrypted admin page or camera stream - HTTP, RTSP

Low

Encrypted services - SSH, HTTPS, MQTT over TLS

None

No risky ports found

Security Recommendations

Telnet	Disable Telnet and use SSH if remote administration is required.
HTTP	Use HTTPS for admin access and change default passwords.
MQTT	Enable authentication and use MQTT over TLS.
RTSP	Enable authentication and restrict camera-stream access.
FTP	Disable FTP and use a secure option like SFTP.

Output - Dashboard

IoT DEVICE SECURITY SCANNER

Network:

127.0.0.1

SCAN NETWORK

Only scan networks you own or have permission to test.

Scan Status: Completed

Devices Found: 1

IoT Devices: 1

Security Findings: 4

DEVICE LIST

Click on a device to see details.

IP Address	Device	Open Ports	Risk
127.0.0.1	IP Camera	554, 1883, 2323, 8080	Critical

- Enter the network range and click Scan Network.
- The page shows devices found, IoT devices and security findings.
- The device list shows IP, device type, open ports and risk.

Output - Device Details

Device Details

IP Address	127.0.0.1
MAC Address	Unknown
Hostname	localhost
Vendor	Unknown
Device Type	IP Camera

Device type is guessed from vendor, hostname and ports. It may not always be correct.

Open Ports

Port	Service	Risk	Reason
554	RTSP	Medium	RTSP is used for camera video streams. Many cameras allow viewing without a password.
1883	MQTT	High	MQTT on this port is not encrypted. Sensor data and device commands can be read or sent by others if authentication is not enabled.
2323	Telnet (alternate)	High	Alternate Telnet port. It gives unencrypted remote access and is commonly targeted by IoT malware such as Mirai.
8080	HTTP (alternate)	Medium	Alternate web admin page without encryption, common on cameras and routers.

Security Recommendations

- **RTSP (554):** Enable authentication and restrict camera-stream access.
- **MQTT (1883):** Enable authentication and use MQTT over TLS where supported.
- **Telnet (alternate) (2323):** Disable Telnet and use SSH if remote administration is required.
- **HTTP (alternate) (8080):** Use HTTPS for administrative access and change default credentials.

- Tested using a fake IoT device running on 127.0.0.1.
- Open ports found: 554, 1883, 2323, 8080.
- Shown as IP Camera with Critical risk.
- Each port has a reason and a recommendation.

Limitations and Future Scope

Limitations

- Only the ports in the list are checked
- Device type is only a guess
- Phones use random MAC addresses, so vendor may be unknown

Future Scope

- Check more ports and services
- Alert when a new device joins the network
- Save scan reports

Conclusion

- The scanner finds devices on a local network and checks them for insecure services.
- Many IoT devices have Telnet, HTTP or unencrypted MQTT open, which can be misused by attackers.
- Turning off unused services, changing default passwords and using SSH, HTTPS and MQTT over TLS improves security.
- Only scan networks you own or have permission to test.

Thank You